RIG-SURJECTIVITY DESCENDS ALONG ETALE COVERS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof assembles base-change stability, etale covers being rig-surjective, composition and the factorization permanence lemma into the descent statement

Editorial note. The label lemma-upshot names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- base change of rig-surjective morphisms (lemma-base-change-rig-surjective of the same chapter)
- etale covers are rig-surjective (lemma-etale-covering-rig-surjective of the same chapter) and rig-surjectivity composes (lemma-composition-rig-surjective of the same chapter)
- permanence of rig-surjectivity along factorizations (lemma-permanence-rig-surjective of the same chapter)

Source reference. Original reference 0AQS: sources/stacks-restricted/pinned-original-source.tex, source lines 5324–5324. Prerequisite pointer invoked at proof line(s) 5406; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQR: sources/stacks-restricted/pinned-original-source.tex, source lines 5307–5307. Prerequisite pointer invoked at proof line(s) 5412; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQU: sources/stacks-restricted/pinned-original-source.tex, source lines 5363–5363. Prerequisite pointer invoked at proof line(s) 5409; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQT: sources/stacks-restricted/pinned-original-source.tex, source lines 5351–5351. Prerequisite pointer invoked at proof line(s) 5414; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally Noetherian formal algebraic spaces which is locally of finite type. Let $\{g_i : Y_i \to Y\}$ be a family of morphisms of formal algebraic spaces which are representable by algebraic spaces and étale such that $\coprod g_i$ is surjective. Then f is rig-surjective if and only if each $f_i : X \times_Y Y_i \to Y_i$ is rig-surjective.*

Proof. Namely, if f is rig-surjective, so is any base change (Lemma 0AQS). Conversely, if all f_i are rig-surjective, so is $\coprod f_i : \coprod X \times_Y Y_i \to \coprod Y_i$. By Lemma 0AQU the morphism $\coprod g_i : \coprod Y_i \to Y$ is rig-surjective. Hence $\coprod X \times_Y Y_i \to Y$ is rig-surjective (Lemma 0AQR). Since this morphism factors through $X \to Y$ we see that $X \to Y$ is rig-surjective by Lemma 0AQT. □